

CSE1021 PROJECT

Indian Income Tax Calculator

A simple Python project

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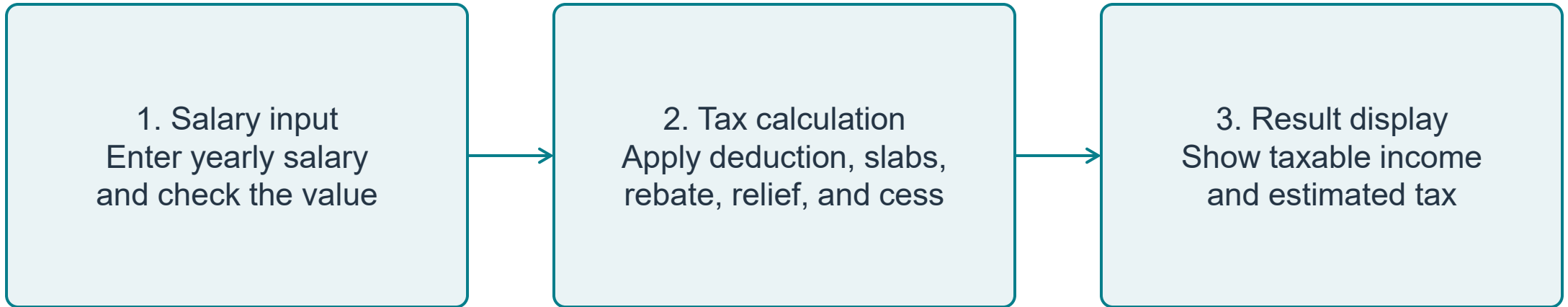
Instructor: Dr. D Lakshmi

Course code: CSE1021

Introduction and Problem

- Tax is calculated in different income bands called slabs.
- Doing the calculation by hand can be confusing.
- This program asks for a salary and gives a simple tax estimate.
- It is a learning project, not a tax filing tool.

Functional Requirements



The user enters one salary. The program calculates and displays the result.

Non-functional Requirements

- Easy to use: prompts and results use simple words.
- Reliable: checks for invalid or negative salary input.
- Fast: calculates one result in a short time.
- Private: does not save the salary or personal details.
- Easy to maintain: comments explain the main calculation steps.

Tax Rules Used

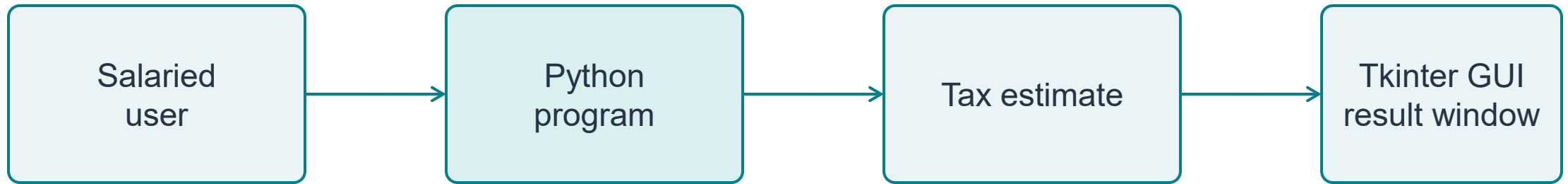
The program uses the new tax regime for FY 2025-26 (AY 2026-27).

- Up to ₹4 lakh: 0%
- ₹4–8 lakh: 5%
- ₹8–12 lakh: 10%
- ₹12–16 lakh: 15%
- ₹16–20 lakh: 20%
- ₹20–24 lakh: 25%
- Above ₹24 lakh: 30%

Salary deduction: up to ₹75,000 | Cess: 4% | Rebate up to ₹12 lakh taxable income for eligible residents

System Architecture

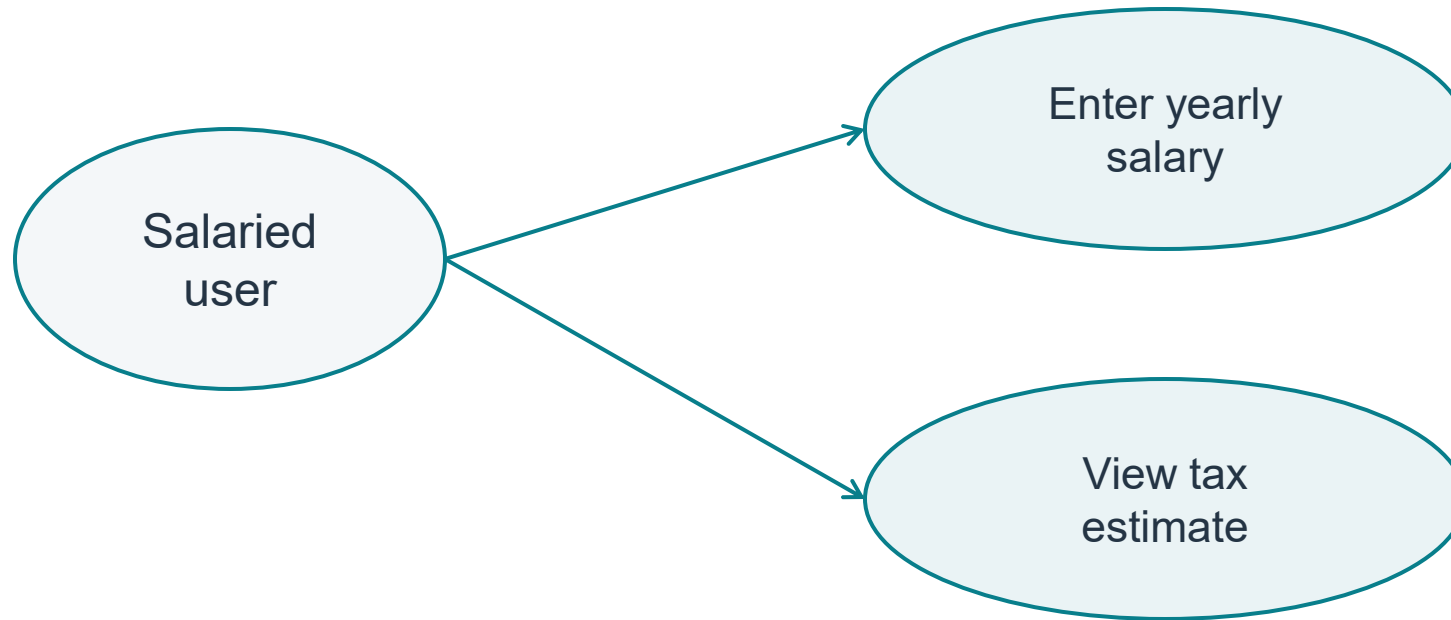
The program works in a simple sequence:



Input salary → calculate tax → show the result

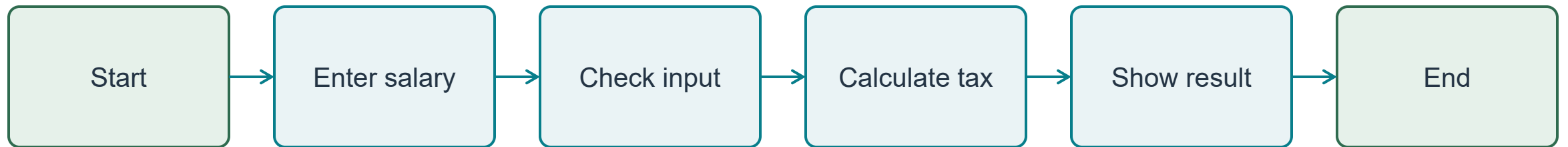
The program does not use a database or save salary details.

Use Case Diagram



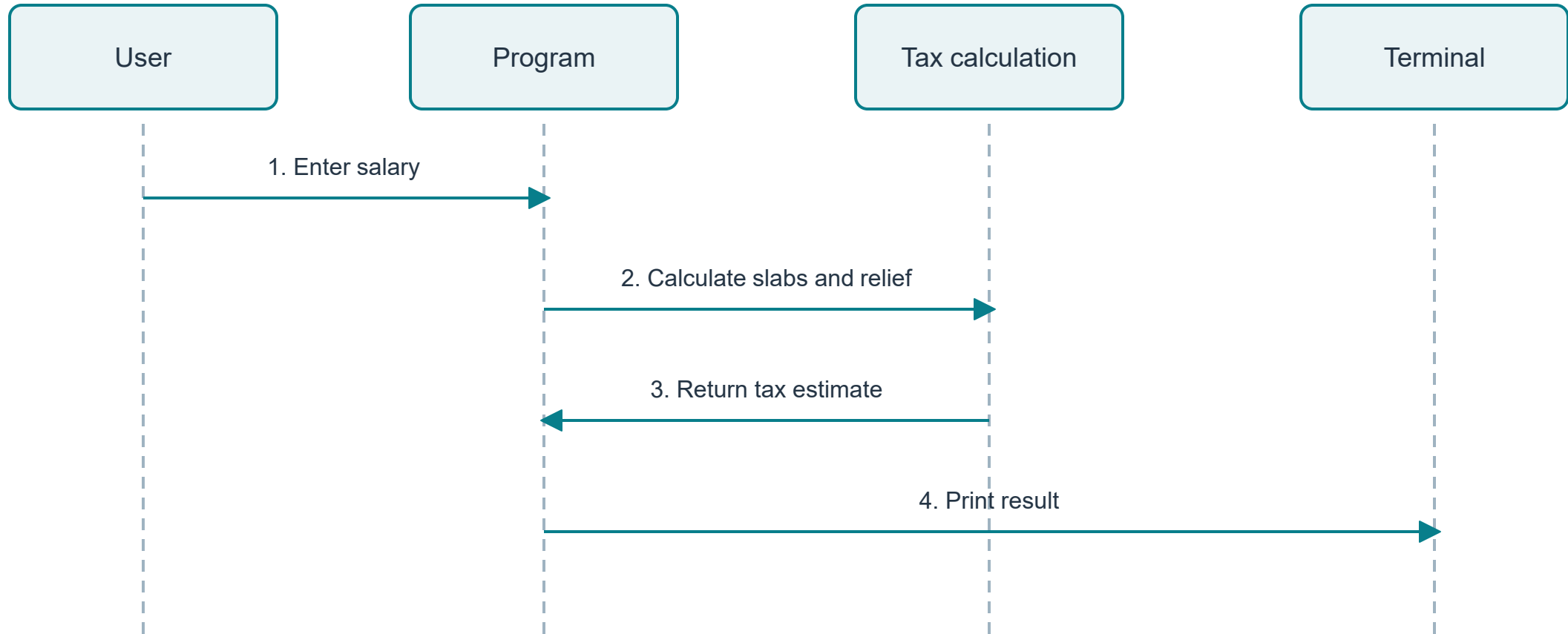
User goal: get an estimated tax result.

Workflow Diagram



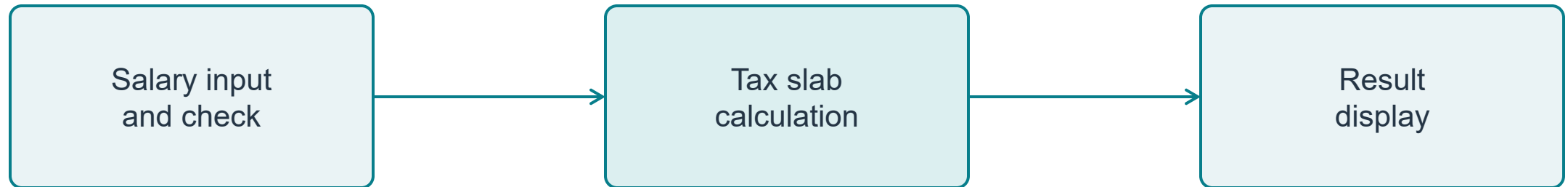
If the input is not a valid non-negative number, the program shows an error and stops.

Sequence Diagram



Program Parts and Data Storage

Main parts inside the Python program



All parts are in the beginner's Python program.

No database is used, so an ER diagram is not needed.

Design Decisions

- Use Tkinter for a graphical interface
- Use progressive slabs because each rate applies to part of the income.
- Use `min()` to stop each slab at its upper limit.
- Do not save salary data because the program only needs one calculation.
- Show the standard deduction and taxable income so the result is easier to follow.

Introduction to Tkinter

- Tkinter is Python's standard library for creating graphical user interfaces (GUIs).
- It allows users to interact with the program through windows, buttons, labels, and input fields.
- The project uses Tkinter to make income tax calculation easier and more user-friendly.
- Users can calculate tax without entering commands in the terminal.

GUI Components Used

Component	Purpose
Tk()	Creates the main window
Label	Displays headings and results
Entry	Accepts annual salary
Button	Performs calculation and exit
messagebox	Displays input errors
StringVar()	Updates calculation results

Implementation Details

Python ideas used

- `input()` and `float()`
- Variables
- if statements
- Arithmetic
- `min()`
- Formatted output
- Tkinter

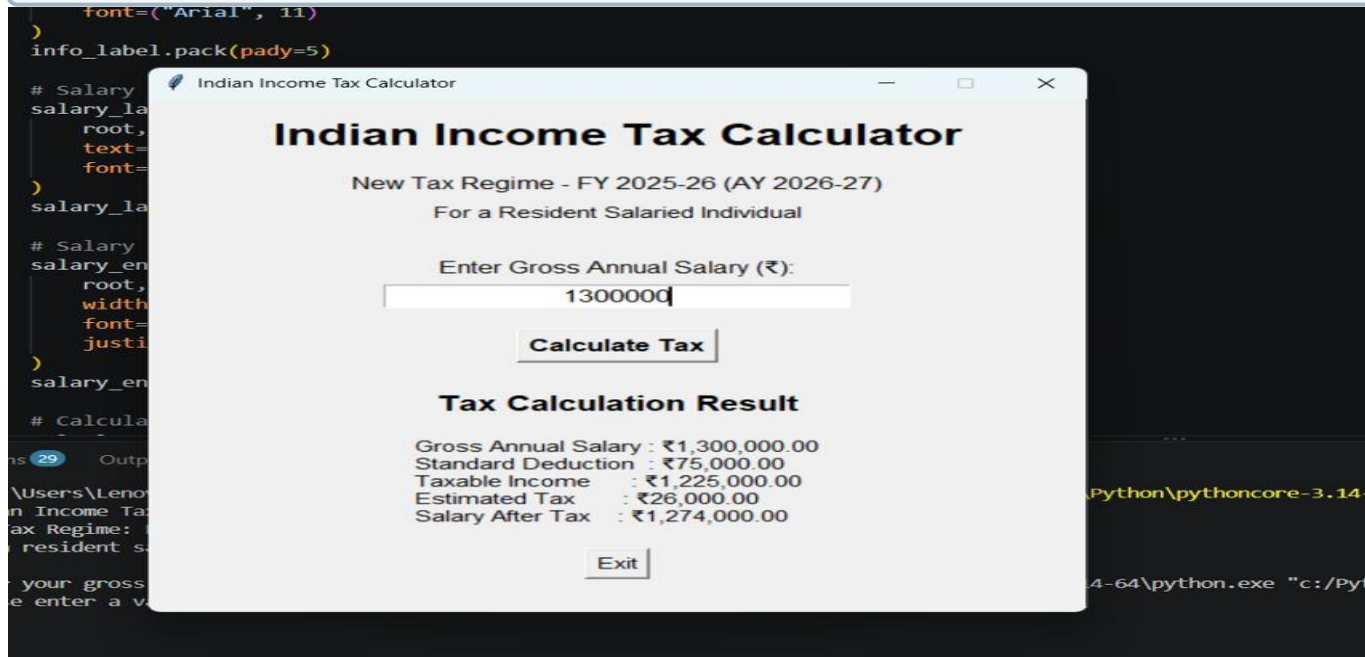
Run the program

1. Open the folder in VS Code.
2. Open Terminal → New Terminal.
3. Type: `python code.py`
4. Enter salary in rupees.

Screenshots and Sample Results

Example: gross salary ₹13,00,000

Gross salary	₹13,00,000
Standard deduction	₹75,000
Taxable income	₹12,25,000
Tax after marginal relief	₹25,000
Cess (4%)	₹1,000
Estimated total tax	₹26,000



Testing Approach and Challenges

Suggested checks (run them in VS Code)

₹12,75,000 salary → expected tax ₹0
₹13,00,000 salary → expected tax ₹26,000
₹15,00,000 salary → expected tax ₹97,500
Text input or negative value → show an error

Main challenges

- Applying each slab to the correct part of income.
- Subtracting the deduction before tax.
- Understanding rebate and marginal relief.

Learnings and Future Enhancements

What I learned

- Read user input
- Use variables and calculations
- Apply if statements
- Print clear results

Possible future changes

- Add more types of income
- Add automatic tests
- Update tax rules for a later year

References

Official sources used for the tax rules

1. Income Tax Department, Government of India
Budget 2025 FAQs: slabs, rebate, marginal relief, and examples
incometaxindia.gov.in/Documents/Budget/budget-2025/faqs-budget-2025.pdf
2. Income Tax Department
Salaried Individuals for AY 2026-27
incometax.gov.in/iec/foportal/help/individual/return-applicable-1

Tax rules can change. This project is an educational estimate only.